8.4 Taxonomic Systems

Classical Taxonomy

Classical taxonomy is a method of classification based on morphology, physiology, biochemical and ecological characteristics of the microorganisms.

- **Morphological Characteristics:** The structural characteristics are the usual tools which help in classification. Cell morphology gives little information about phylogenetic relationship. The first step in identification of bacteria is differential staining.
- **Physiological and metabolic characteristics:** These characteristics are useful because they are directly related to nature and activity of microbial enzymes and transport protein. Since proteins are gene products, analysis of these characteristics provides an indirect comparison of microbial genomes.
- **Biochemical characteristics:** Enzymatic activities are widely used to differentiate bacteria. Bacteria can be separated into separate species by various biochemical tests. Example: Carbohydrate fermentation ability of bacteria.
- **Ecological characteristics:** Many properties are ecological in nature since they alter the relation of microorganism to their environment. Microorganisms living in various parts of the human body markedly differ from one another and from those growing in freshwater, terrestrial and marine environments.

Prokaryotes have only a few structural characteristics and these characteristics are subject to rapid change due to change in environment. In classifying prokaryotes, metabolic reactions, genetic relatedness and other specialized properties are used (Figure 8.4).

HOTS

If two microorganisms have an identical mol% G+C value for their DNA, are they necessarily related? Explain

If two micro organisms have very different mol% G+C values for their DNA, are they necessarily unrelated? Explain

Numerical Taxonomy

The objective classification system deals with the grouping by numerical methods of taxonomic units based on their character and does not use subjective evaluation of their properties. To be more objective about grouping bacteria, the scientists determine many characteristics (usually 100 to 200) for each strain studied, giving equal weightage for each character. Then by using computer **%similarity** is calculated **(%S** of each strain to every other strain). For any two strains, this is

$$\%S = \frac{NS}{NS + ND}$$

where, NS is the number of characteristics that are the same (positive or negative) for the two strains, and ND is the number of characteristics that are different. Those strains having a high %S to each other are placed into groups; and those groups having a high %S to each other are in turn placed into larger

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